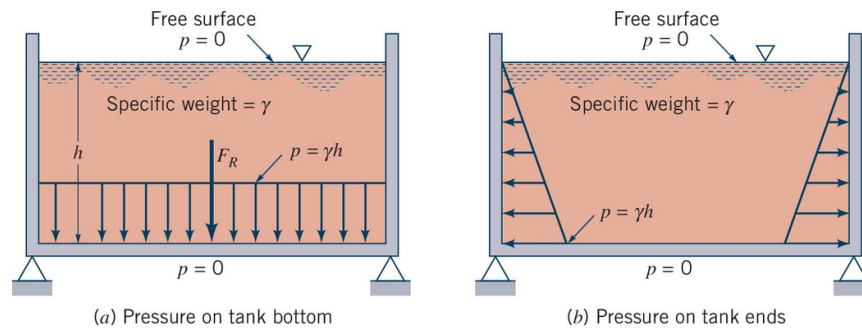


2.8 Hydrostatic force on a plane surface

For fluid at rest, fluid pressure acts perpendicular to any surface because there is no shear stress present in a static fluid.

For example, the pressure and force distribution on an open tank is shown below:



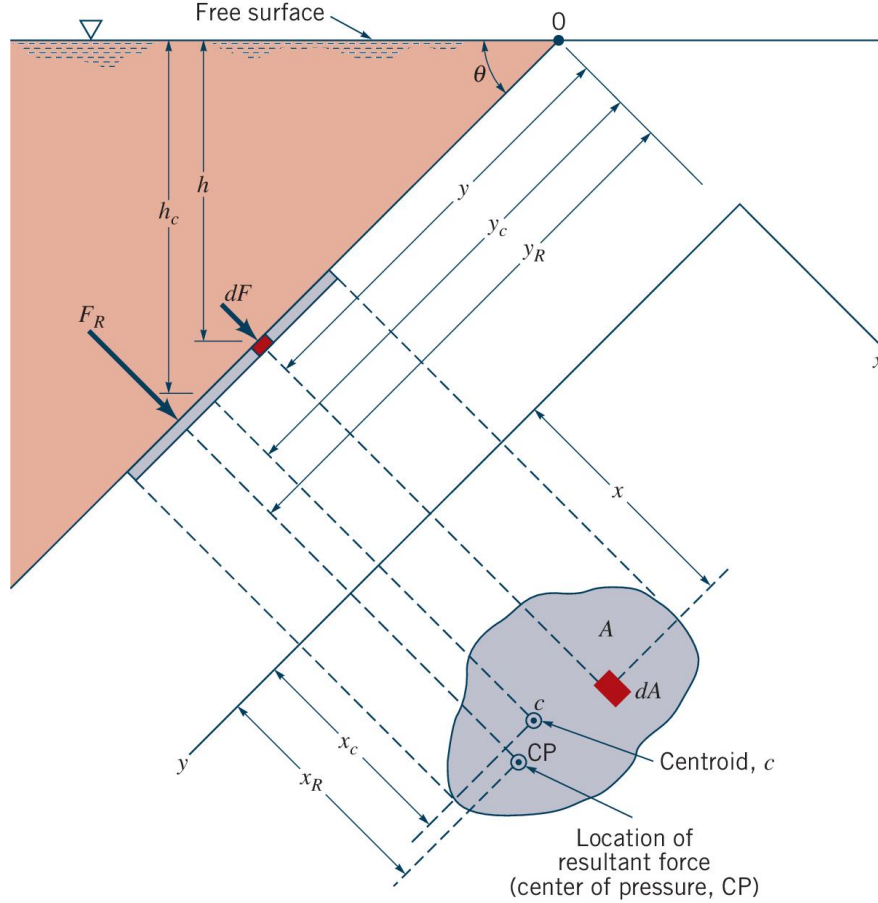
Note that if atmospheric pressure acts on both sides and bottom of the tank, it will cancel out and can be ignored (which is why we often use gage pressure). The resultant force on the bottom of tank is given by:

$$F_R = pA, \quad \text{where } p = \gamma h$$

Since the pressure is constant and uniformly distributed over the bottom, the resultant force acts through the **centroid of the area**. However, for the sides of the tank the pressure varies with depth, so it's harder to determine the resultant force and its line of action.

Resultant Force on a Plane Surface

Consider the following submerged inclined plane surface below. The free surface is open to the atmosphere. We define a xy coordinate system so that the surface lies on the xy plane, with the origin where the plane in which the surface lies intersect the free surface.



The force due to the fluid pressure in contact with the surface is given by integrating the pressure over the area of the surface:

$$F_R = \int_A p dA = \int_A \gamma h dA$$

Where h is the depth of the surface element dA below the free surface (h is in the vertical direction, NOT in the y direction). Note that the depth h can be expressed in terms of the y coordinate by the angle of inclination θ :

$$h = y \sin \theta$$

Also note that the integral for F_R can be expressed in terms of the first moment of area of the surface about the x -axis:

$$Q = \int_A y dA = A\bar{y} \quad (\text{First Moment of Area})$$

Where y is the distance from the x -axis to the area element dA , and $\bar{y}$ is the distance from the x -axis to the centroid of the area. Thus, using some new notation, we can express the resultant force as:

$$F_R = \int_A \gamma h dA = \int_A \gamma y \sin \theta dA$$

For a constant γ and θ , and $h = y \sin \theta$:

$$F_R = \gamma \sin \theta \int_A y dA = \gamma \sin \theta A y_c$$

Where $\bar{y} = y_c$ is the distance from the x -axis to the centroid of the area along the y -axis. Finally:

$$\boxed{F_R = \gamma h_c A}$$

y Coordinate of the Resultant Force

Recall that the resultant force acts through the centroid of the area when the pressure is constant (like the bottom of the tank). However, when the plane surface is inclined, the pressure increases with depth, and the resultant force acts slightly below the centroid of the area.

We can determine the y coordinate of the resultant force (y_R) by summing moments about the x -axis. The moment of the resultant force at y_R is equal to the moment of the distributed pressure about the x -axis:

$$\begin{aligned} F_R y_R &= \int_A y dF, \quad dF = \gamma h dA = \gamma y \sin \theta dA \\ \implies F_R y_R &= \int_A \gamma \sin \theta y^2 dA \end{aligned}$$

From before, $F_R = \gamma h_c A = \gamma \sin \theta A y_c$. So for a constant γ and θ :

$$\begin{aligned} \gamma \sin \theta A y_c y_R &= \gamma \sin \theta \int_A y^2 dA \\ \implies y_R &= \frac{\int_A y^2 dA}{y_c A} \end{aligned}$$

Note that the integral in the numerator is the second moment of area about the x -axis, which is also known as the *area moment of inertia*:

$$I_x = \int_A y^2 dA$$

Thus, we can express the y coordinate of the resultant force as:

$$y_R = \frac{I_x}{y_c A}$$

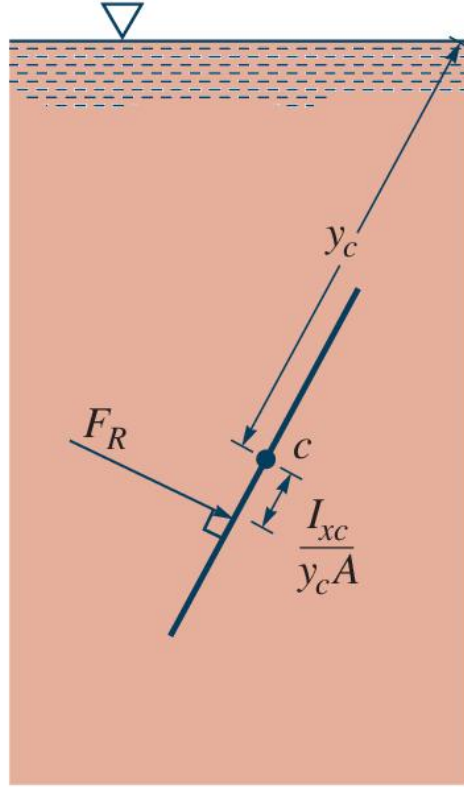
Using the parallel axis theorem, we can express I_x in terms of an axis parallel to the x -axis that passes through the centroid of the area:

$$I_x = I_{xc} + Ay_c^2$$

Thus,

$$y_R = \frac{I_{xc}}{y_c A} + y_c$$

Following the intuition from the beginning, since $I_{xc}/(y_c A)$ is always positive, the resultant force will always act below the centroid of the area.



Note: y_c is measured from the line of intersection of the xy plane with the surface, defining the angle θ . However, when the xy plane is horizontal and parallel to the free surface, $\theta \rightarrow 0^\circ$, and $y_c = h_c / \sin \theta \rightarrow \infty$. In this case, the term $I_{xc}/(y_c A) \rightarrow 0$, and the resultant force acts through the centroid of the area, as expected.

x Coordinate of the Resultant Force

Similarly, we can determine the x coordinate of the resultant force (x_R) by summing moments about the y -axis. The moment of the resultant force at x_R is equal to the moment of the distributed pressure about the y -axis:

$$F_R x_R = \int_A x dF, \quad dF = \gamma h dA = \gamma y \sin \theta dA$$

$$\implies F_R x_R = \int_A \gamma \sin \theta xy dA$$

Since $F_R = \gamma h_c A = \gamma \sin \theta A y_c$, for a constant γ and θ , we can divide through by F_R to get:

$$x_R = \frac{\gamma \sin \theta \int_A xy dA}{\gamma \sin \theta A y_c} = \frac{I_{xy}}{y_c A}$$

Like before, I_{xy} is the product of inertia about the x and y axes. Using the parallel axis theorem, we get the general expression for the x coordinate of the resultant force:

$$x_R = \frac{I_{xyc}}{y_c A} + x_c$$

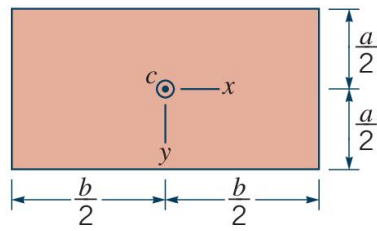
Where I_{xyc} is the product of inertia about the x and y axes that pass through the centroid of the area, and x_c is the x coordinate of the centroid of the area.

Note that if the plane surface is symmetrical about an axis parallel to either the x or y axis and passing through the centroid of the area, then $I_{xyc} = 0$, and $x_R = x_c$.

Together, x_R and y_R define the *center of pressure*, which is the point where the resultant force acts on the plane surface.

Note: For both x_R and y_R , the center of pressure will move closer to the centroid of the area as the plane surface becomes more horizontal ($\theta \downarrow$), and thus increases y_c in the denominator of both expressions. In the limit as $\theta \rightarrow 0^\circ$, $y_c \rightarrow \infty$, and the center of pressure coincides with the centroid of the area, as expected.

Common Area Moments of Inertia



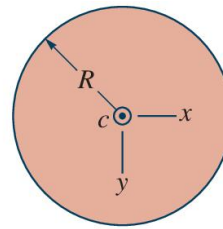
(a) Rectangle

$$A = ba$$

$$I_{xc} = \frac{1}{12} ba^3$$

$$I_{yc} = \frac{1}{12} ab^3$$

$$I_{xyc} = 0$$

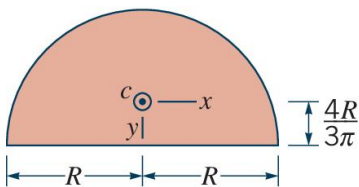


(b) Circle

$$A = \pi R^2$$

$$I_{xc} = I_{yc} = \frac{\pi R^4}{4}$$

$$I_{xyc} = 0$$



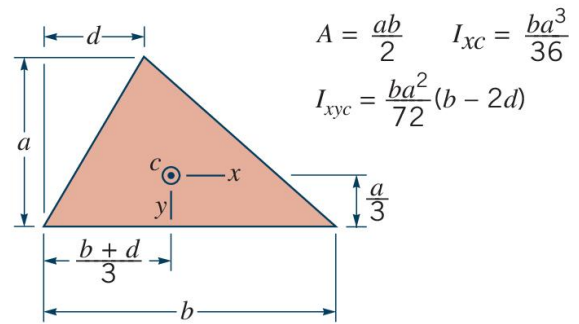
(c) Semicircle

$$A = \frac{\pi R^2}{2}$$

$$I_{xc} = 0.1098R^4$$

$$I_{yc} = 0.3927R^4$$

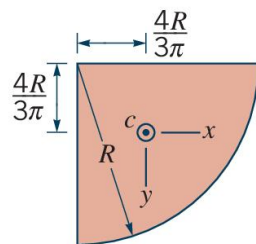
$$I_{xyc} = 0$$



(d) Triangle

$$A = \frac{ab}{2} \quad I_{xc} = \frac{ba^3}{36}$$

$$I_{xyc} = \frac{ba^2}{72}(b - 2d)$$



(e) Quarter circle

$$A = \frac{\pi R^2}{4}$$

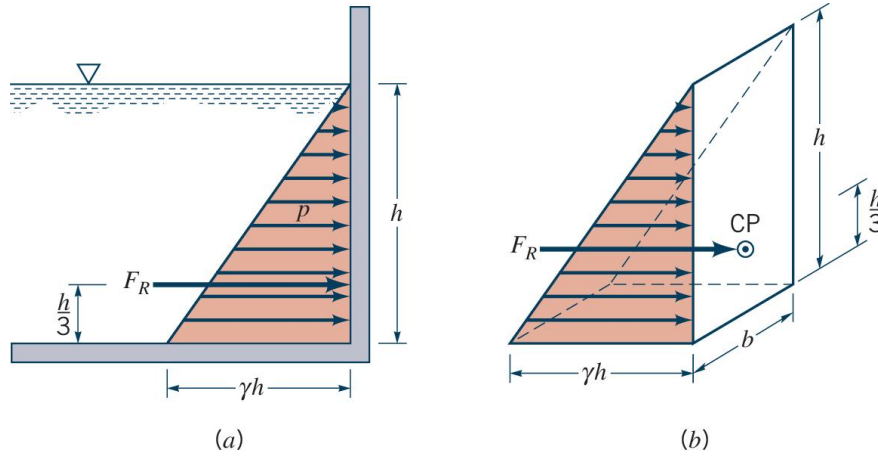
$$I_{xc} = I_{yc} = 0.05488R^4$$

$$I_{xyc} = -0.01647R^4$$

2.9 Pressure Prism

This section doesn't introduce any new concepts, but rather provides a graphical approach to solving for the resultant force and center of pressure on a plane surface. This approach is often convenient for rectangular surfaces, where the centroid and volume of the pressure prism can be easily determined.

Consider the following pressure prism on a vertical plane surface with constant width b :



Since the pressure varies linearly with depth, it is zero at the free surface ($p_{min} = 0$) and maximum at the bottom of the surface ($p_{max} = \gamma h$). Thus, the average pressure on the surface is $p_{avg} = \frac{p_{min} + p_{max}}{2} = \frac{\gamma h}{2}$. The resultant force on the surface is given by:

$$F_R = p_{avg}A = \gamma \left(\frac{h}{2} \right) A$$

This is the same result as the equation $F_R = \gamma h_c A$ in section 2.8.

By forming a *pressure prism* (shown in figure (b)), the resultant force can be determined graphically as the **volume of the pressure prism**:

$$F_R = \text{volume} = \frac{1}{2}(\gamma h)(bh)$$

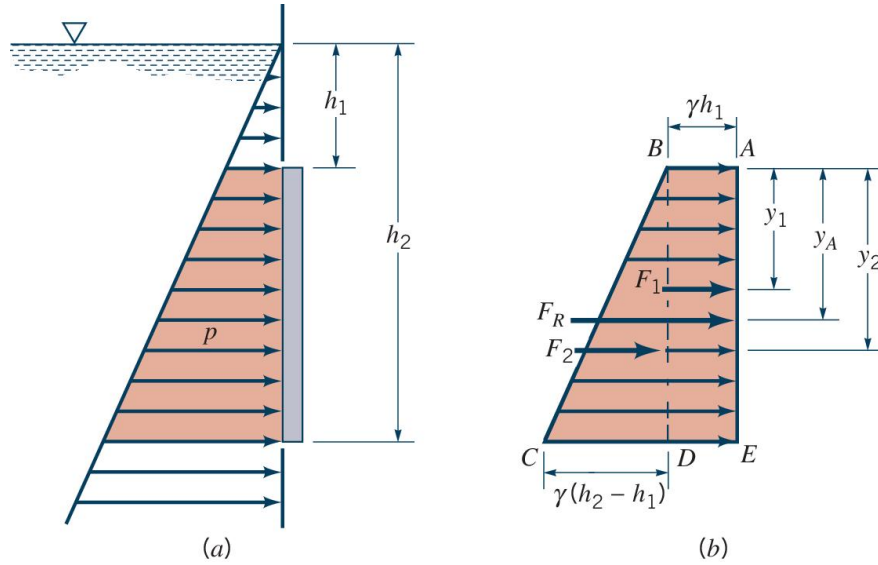
This is the same result as before as well.

The center of pressure can also be determined graphically as the **centroid of the pressure prism**. The centroid of a triangle is located at a distance of $\frac{1}{3}$ from the base, so the center of pressure is located at a distance of $\frac{h}{3}$ from the bottom of the surface. This is the same result as the equation $y_R = I_{xc}/(y_c A) + y_c$ in section 2.8.

Note: The pressure prism is always drawn starting from the free surface.

Rectangular Surfaces that do not intersect the free surface

Consider the following plane surface that does not intersect the free surface:



The pressure prism approach can still be used but in this case, it is trapezoidal instead of triangular. The resultant force can be found by decomposing the trapezoid into a rectangle and triangle, $ABDE$ and BCD , respectively.

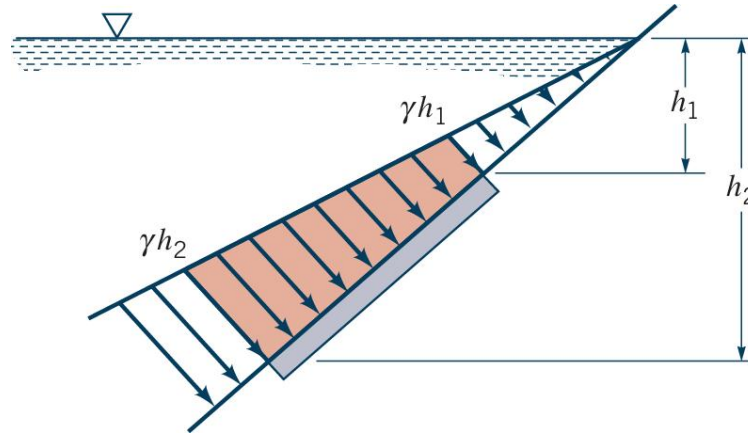
$$F_R = F_1 + F_2$$

The locations of the forces F_1 and F_2 can be found by taking the centroid of each shape, getting y_1 and y_2 , respectively. The location of the resultant force can then be found by summing moments about a convenient axis (in this case, the axis passing through A):

$$F_R y_R = F_1 y_1 + F_2 y_2$$

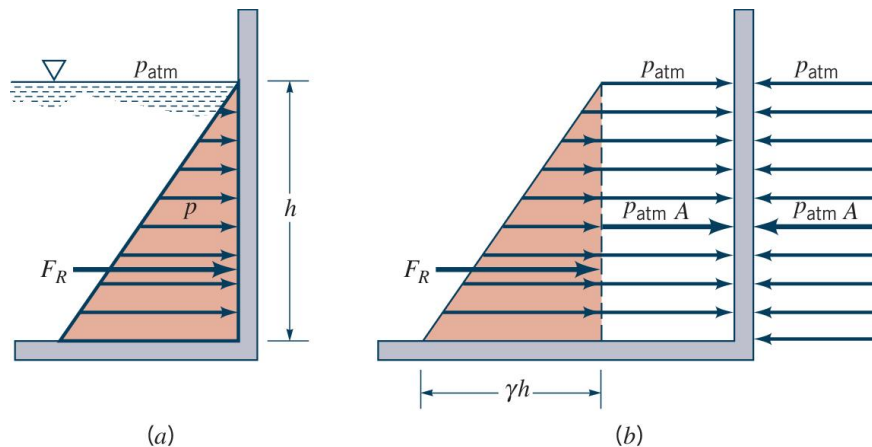
Inclined Rectangular Surfaces

A similar approach using trapezoidal pressure prisms can be used for inclined rectangular surfaces. However, note that while it is convenient to measure distances along the incline, the pressure varies with depth in the vertical direction.



Effect of Pressure at Free Surface

When we consider the effect of atmospheric pressure at the free surface, it simply offsets the pressure prism by a constant amount, resulting in a trapezoidal pressure prism.



In cases of open tanks, the atmospheric pressure acts on both sides of the vertical walls and so it cancels out. However, in cases where the pressure at the free surface is not equal to the pressure on the exterior surface, its effect is non-negligible.

Anyway, the analysis follows the same approach as the trapezoidal pressure prism.

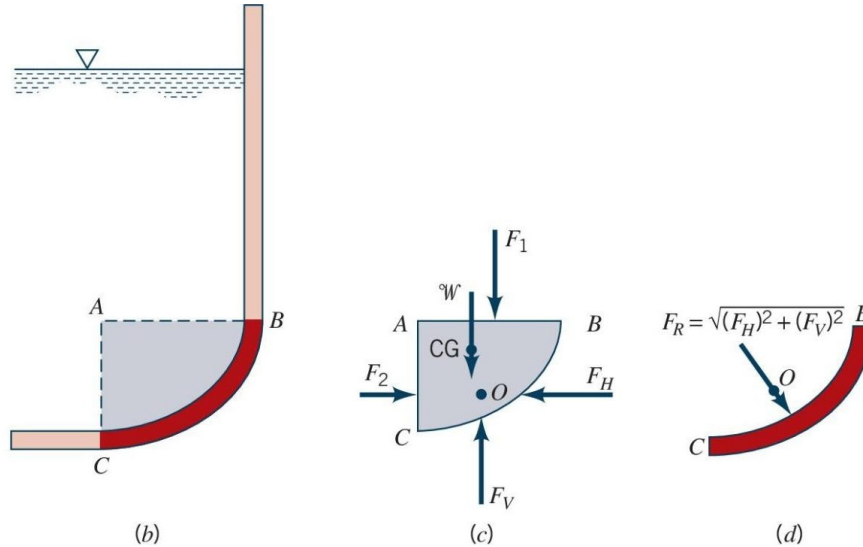
In summary, the volume of the pressure prism is basically:

$$(\text{Average pressure}) \cdot (\text{Area in contact with fluid})$$

2.10 Hydrostatic force on a curved surface

So far, we have only considered plane surfaces. What about curved surfaces? The resultant force on a curved surface can be determined by decomposing the force into horizontal and vertical components.

For example, consider the following curved surface BC :



Start by isolating a volume of fluid that is bounded by the surface of interest (b). Next, we determine the forces in the free-body diagram of the isolated fluid (c).

- F_1 : Force by fluid pressure on the *planar surface* AB . This can be determined using the methods from sections 2.8 and 2.9.
- F_2 : Force by fluid pressure on the *planar surface* AC . This can also be determined using the methods from sections 2.8 and 2.9.
- W : **Weight of the enclosed fluid volume**. This can be determined by multiplying the volume of the fluid by its specific weight and acts through the center of gravity of the volume.
- F_H and F_V : **The horizontal and vertical reaction forces from the tank on the fluid.**

Since the fluid is at static equilibrium, the sum of forces in both the horizontal and vertical directions must be zero.

$$\sum F_{\text{horiz}} = 0 = F_2 - F_H \implies F_H = F_2$$

$$\sum F_{\text{vert}} = 0 = F_1 + W - F_V \implies F_V = F_1 + W$$

Additionally, **for a body under three non-parallel forces to be in equilibrium, the lines of action of the three forces must intersect at a common point** (otherwise there would be a

non-zero moment). Therefore, we can say that the resultant force F_R passes through the common intersection point of the lines of action of F_H and F_V , O .

$$F_R = \sqrt{F_H^2 + F_V^2}$$

O can be determined by summing moments to find the lines of action of F_H and F_V , and then finding their intersection.

Note: The vertical component of the resultant force can actually be solved two ways:

1. Instead of isolating a small fluid region around the surface, we can isolate the entire fluid column above it. Then, the vertical hydrostatic force on the surface is just the weight of the fluid column above it:

$$F_V = \mathcal{W} = \gamma \mathcal{V}$$

2. Or alternatively, the procedure described before can be used, where we isolate a local fluid region around the surface. In that case, the vertical hydrostatic force on the surface is the sum of the weight of the fluid region and the vertical component of the hydrostatic force:

$$F_V = F_1 + \mathcal{W}$$

2.11 Buoyancy, flotation, and stability

Archimedes' Principle

The buoyant force on a body submerged in a fluid is equal to the weight of the fluid displaced by the body.

$$F_{\text{Buoyant}} = \gamma \mathcal{V}$$

Line of Action for the Buoyant Force

The point through which the buoyant force acts is called the *center of buoyancy*. The buoyant force passes through the centroid of the displaced volume.

For partially submerged bodies, if the specific weight of the fluid above the liquid surface is very small compared to the displaced fluid, the center of buoyancy can be approximated as the centroid of the submerged volume. This is a common approximation when the fluid above the surface is air.

If a body is submerged in a fluid of varying specific weight, such as a layered fluid, the magnitude of the buoyant force remains equal to the weight of the displaced fluid. However, the center of buoyancy is located at the center of gravity of the displaced fluid.

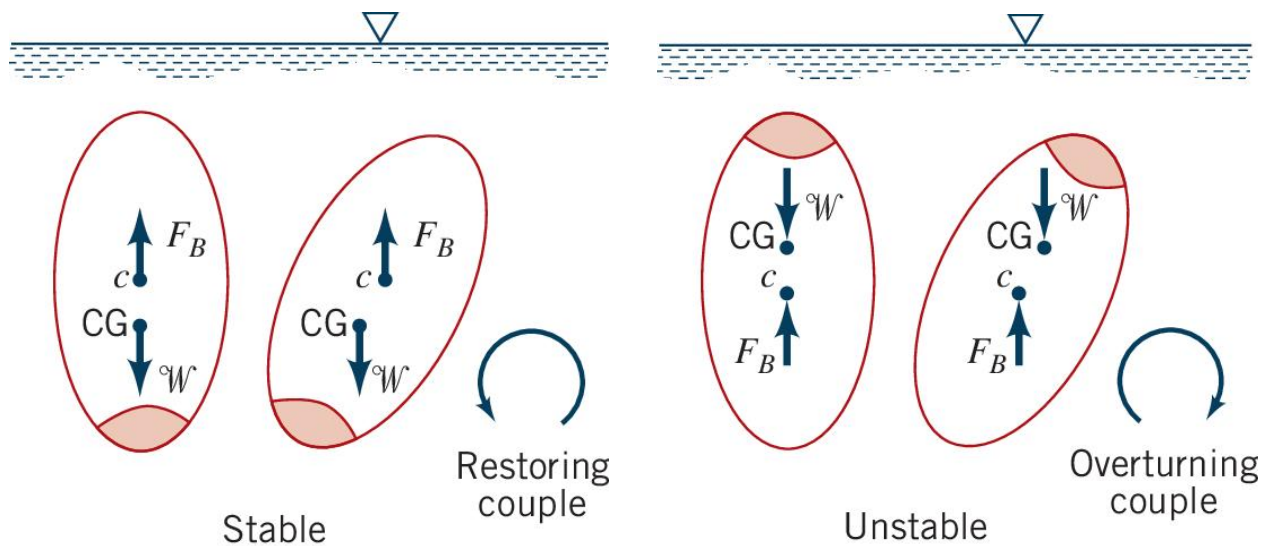
Note: The buoyant force is a special application of the hydrostatic pressure forces acting on the surfaces of a body. In other words, both buoyancy and hydrostatic pressure forces arise from the same physical phenomenon. Therefore, when performing calculations, use

either the resultant buoyant force or the individual hydrostatic pressure forces, but do not include both.

Stability

A body is said to be in stable equilibrium if it returns to its original position after being slightly disturbed, and in unstable equilibrium if it moves to a new equilibrium position instead.

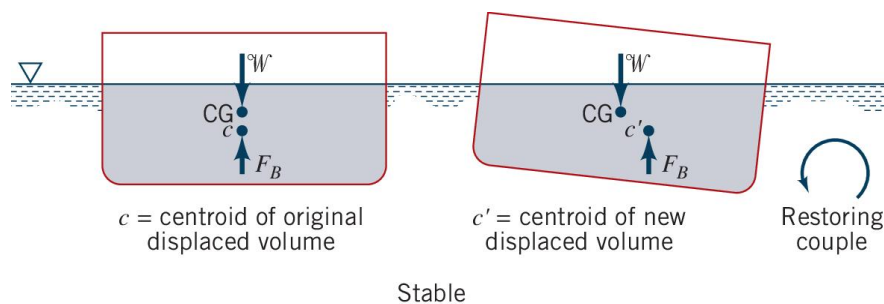
Stability of a **submerged body** can be determined by comparing the center of gravity of the body and the center of buoyancy:



The body is in stable equilibrium if the center of gravity is below the center of buoyancy, and in unstable equilibrium if the center of gravity is above the center of buoyancy.

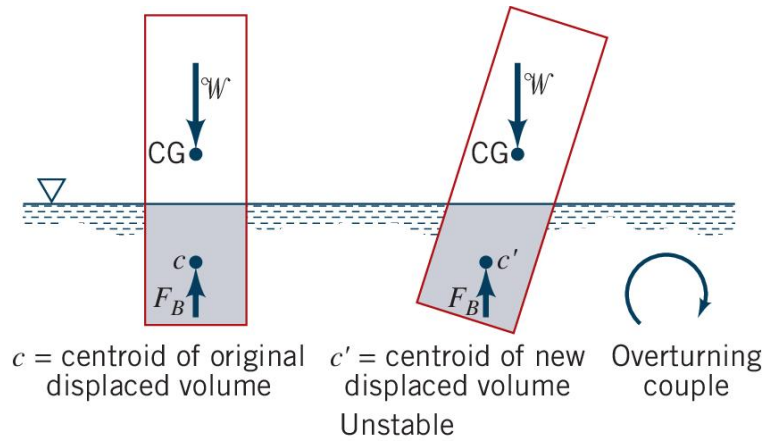
Floating Body Stability

For a floating body, the center of buoyancy (which is approximately the centroid of the submerged volume) *moves as the body is tilted*. Thus, it is possible for a wide floating body to be in stable equilibrium even if its center of gravity is above the center of buoyancy as seen below:



However, a narrow floating body may be in unstable equilibrium if its center of gravity

is above the center of buoyancy since the center of buoyancy may not move enough to counteract the tilt.



2.12 Pressure variation in a fluid with rigid-body motion

Recall that the general equation of motion for a fluid with no shearing stresses is given by:

$$-\nabla p - \gamma \hat{k} = \rho \vec{a}$$

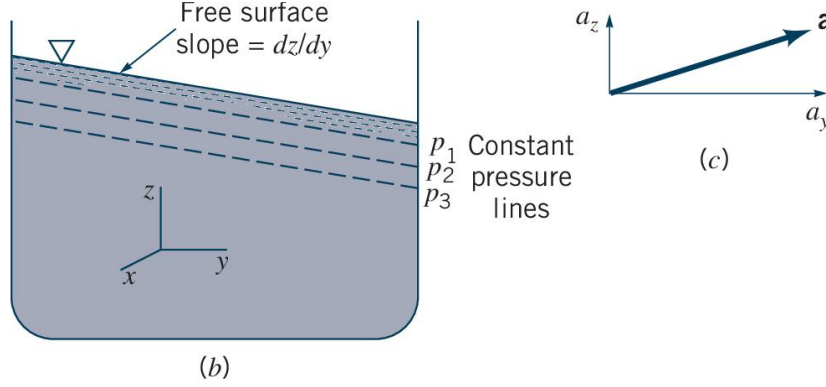
$$\Rightarrow -\frac{\partial p}{\partial x} = \rho a_x, \quad -\frac{\partial p}{\partial y} = \rho a_y, \quad -\frac{\partial p}{\partial z} = \gamma + \rho a_z$$

Problems involving fluids in motion but having no shearing stresses occur when the fluid undergoes rigid-body motion. In this case, the acceleration of all fluid particles is the same (after some transient sloshing dies down). For example, if a container of fluid accelerates along a straight path, the fluid will move as a rigid mass, and the fluid would not deform (i.e., no shearing stresses).

By using the equation of motion for a fluid with no shearing stresses, we can determine the pressure distribution in a fluid in motion.

Linear Motion

Consider an open container of fluid that is moving with a constant acceleration a along a straight path.



Since $a_x = 0$ it implies that $\partial p / \partial x = 0$, and thus the pressure is constant along the x direction. Additionally, in the y and z directions, the pressure distribution is given by:

$$\frac{\partial p}{\partial y} = -\rho a_y$$

$$\frac{\partial p}{\partial z} = -\gamma - \rho a_z = -\rho(g + a_z)$$

The change in pressure between two closely spaced points in the fluid at (x, y, z) and $(x, y + dy, z + dz)$ is given by the first-order Taylor series expansion:

$$dp = \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz$$

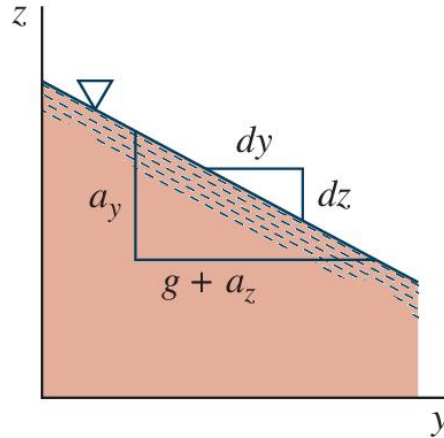
Plugging in the expressions for $\partial p / \partial y$ and $\partial p / \partial z$, we get:

$$dp = -\rho a_y dy - \rho(g + a_z) dz$$

Along a line of **constant pressure**, $dp = 0$ by definition, thus the equation above can be rearranged to give the slope of the constant pressure line:

$$\frac{dz}{dy} = -\frac{a_y}{g + a_z}$$

Note that for any accelerating fluid, $a_y \neq 0$ and thus $dz/dy \neq 0$, which means that the free surface of the fluid will have a non-zero slope. Additionally, ALL lines of constant pressure will be parallel to the free surface (with the same slope).



Vertical Acceleration

For the special case of pure vertical acceleration where $a_y = 0$ and $a_z \neq 0$, the slope of the free surface is zero, and the free surface remains horizontal. However, the pressure distribution is NOT hydrostatic but is given by:

$$\frac{dp}{dz} = -\rho(g + a_z)$$

In other words, the pressure distribution is still linear but now includes a term for the vertical acceleration:

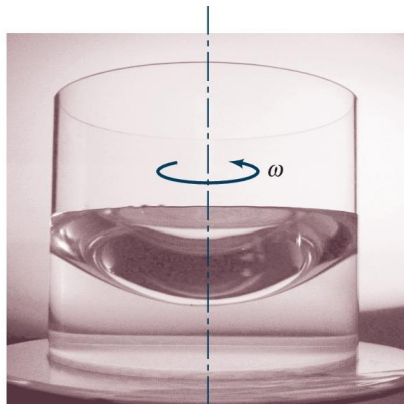
$$p = p_0 - \rho(g + a_z)z$$

Note: Pressure still increases with depth, but here z is defined as positive upwards. So, $z < 0$ below the free surface.

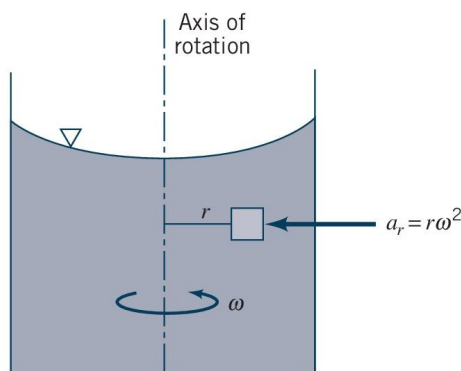
It can also be seen that if the fluid is in free fall, $a_z = -g$, then $dp/dz = 0$, and the pressure is constant throughout the fluid.

Rigid-body rotation

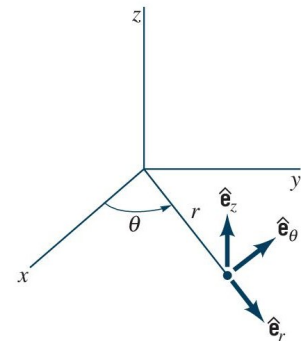
Consider a container of fluid that is rotating about a vertical axis with a constant angular velocity ω .



(a)



(b)



(c)

Since the paths of all fluid particles are circular, the acceleration of a fluid particle a distance r from the axis of rotation is given by $r\omega^2$, and it is more convenient to use cylindrical coordinates (r, θ, z) instead of Cartesian coordinates (x, y, z) . The gradient operator in cylindrical coordinates is given by:

$$\nabla = \frac{\partial}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{e}_\theta + \frac{\partial}{\partial z} \hat{e}_z$$

Thus the equation of motion for a fluid with no shearing stresses in cylindrical coordinates is given by:

$$\begin{aligned} -\nabla p - \gamma \hat{e}_z &= \rho \vec{a} \\ -\frac{\partial p}{\partial r} \hat{e}_r - \frac{1}{r} \frac{\partial p}{\partial \theta} \hat{e}_\theta - \left(\frac{\partial p}{\partial z} + \gamma \right) \hat{e}_z &= \rho a_r \hat{e}_r + \rho a_\theta \hat{e}_\theta + \rho a_z \hat{e}_z \end{aligned}$$

Since the accelerations in the cylindrical coordinate system are given by:

$$\vec{a}_r = -r\omega^2 \hat{e}_r, \quad \vec{a}_\theta = 0, \quad \vec{a}_z = 0$$

Thus, setting components equal:

$$\frac{\partial p}{\partial r} = \rho r \omega^2, \quad \frac{1}{r} \frac{\partial p}{\partial \theta} = 0, \quad \frac{\partial p}{\partial z} = -\gamma$$

Since there are two non-zero components, the pressure distribution is in terms of two variables, r and z . Again with the first order Taylor series expansion, the change in pressure between two closely spaced points in the fluid at (r, θ, z) and $(r + dr, \theta, d\theta, z + dz)$ is given by:

$$dp = \frac{\partial p}{\partial r} dr + \frac{\partial p}{\partial z} dz \implies dp = \rho r \omega^2 dr - \gamma dz$$

For a **surface of constant pressure**, $dp = 0$, and using $\gamma = \rho g$, the slope of the constant pressure surface is given by:

$$\frac{dz}{dr} = \frac{r\omega^2}{g}$$

Integrating this differential equation gives the equation of the free surface of the fluid as a parabolic surface:

$$z(r) = \frac{\omega^2}{2g} r^2 + \text{constant}$$

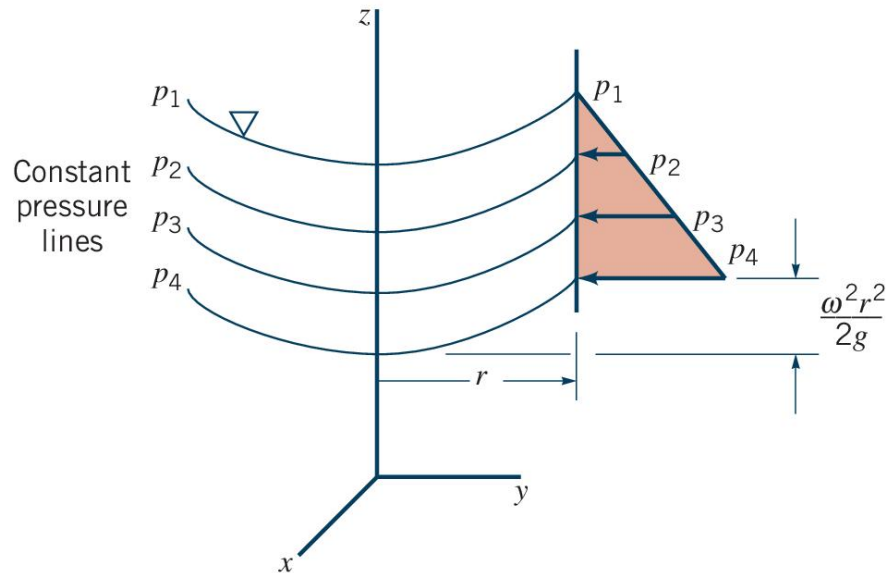
Pressure Distribution

Integrating the equation for dp gives the pressure distribution in the fluid:

$$\begin{aligned} \int dp &= \int \rho r \omega^2 dr - \int \gamma dz \\ \implies p(r, z) &= \frac{1}{2} \rho \omega^2 r^2 - \gamma z + \text{constant} \end{aligned}$$

These relations show that

- The free surface of a fluid in rigid-body rotation is parabolic.
- While the pressure distribution of a fluid in rigid-body rotation depends on both r and z , at a fixed radius r , the pressure distribution is linear with depth z .



Note: Once again note that positive z is defined as upward, so z is negative below the free surface.

Also note that when calculating the pressure *difference*, the constant of integration cancels out. Otherwise, the constant of integration can be determined by applying a boundary condition, such as the pressure at the free surface.